

Derrickson Ymbon

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North-Eastern Hill University
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Research Interests

Quiver Algebra, Representation Theory and Category Theory.

Education

M.Sc. in Mathematics 2024 -
CGPA: 6.5
North-Eastern Hill University

B.Sc. in Mathematics 2021 - 2024
CGPA: 5.5
North-Eastern Hill University

Relevant Courses

Graduate Algebra, Analysis, Linear Algebra, Topology, Differential Geometry.

Projects

- **Master’s Project: Surface Triangulations and the Structures of Gentle and Brauer Graph Algebras**
(February 2026 – present)
Supervisors: Dr. Ardeline Mary Buhphang and Prof. Thomas Brüstle, University of Sherbrooke, Canada.

Study of the relationship between triangulated surfaces and the associated classes of gentle algebras and Brauer graph algebras, focusing on their combinatorial and representation-theoretic structures.

- **Reading Project: Homological Algebra in Representation Theory**
(January 2026 – present) Mentor: Dr. Ardeline Mary Buhphang

Ongoing guided reading, the project follows Henning Krause’s *Homological Theory of Representations* together with C. Weibel’s *An Introduction to Homological Algebra*.

Topics studied include projective and injective modules, projective and injective resolutions, derived functors such as Ext and Tor, chain complexes and chain homotopies, and the construction of homotopy and derived categories. Particular emphasis is placed on the role of these homological tools in the study of module categories of finite-dimensional algebras and in the representation theory of quivers.

- **Reading Project: Quiver Representations and Auslander–Reiten Theory**
(February 2026 – present)
Mentors: Prof. Thomas Brüstle (University of Sherbrooke), Dr. Ardeline Mary Buhphang (NEHU)

Study of quiver representations based on Ralf Schiffler’s *Quiver Representations*, covering Gabriel’s theorem, indecomposable representations, and an introduction to Auslander–Reiten theory.

Computer Skills

- Programming Language: C, Python.
- Typesetting: $\text{\LaTeX}$